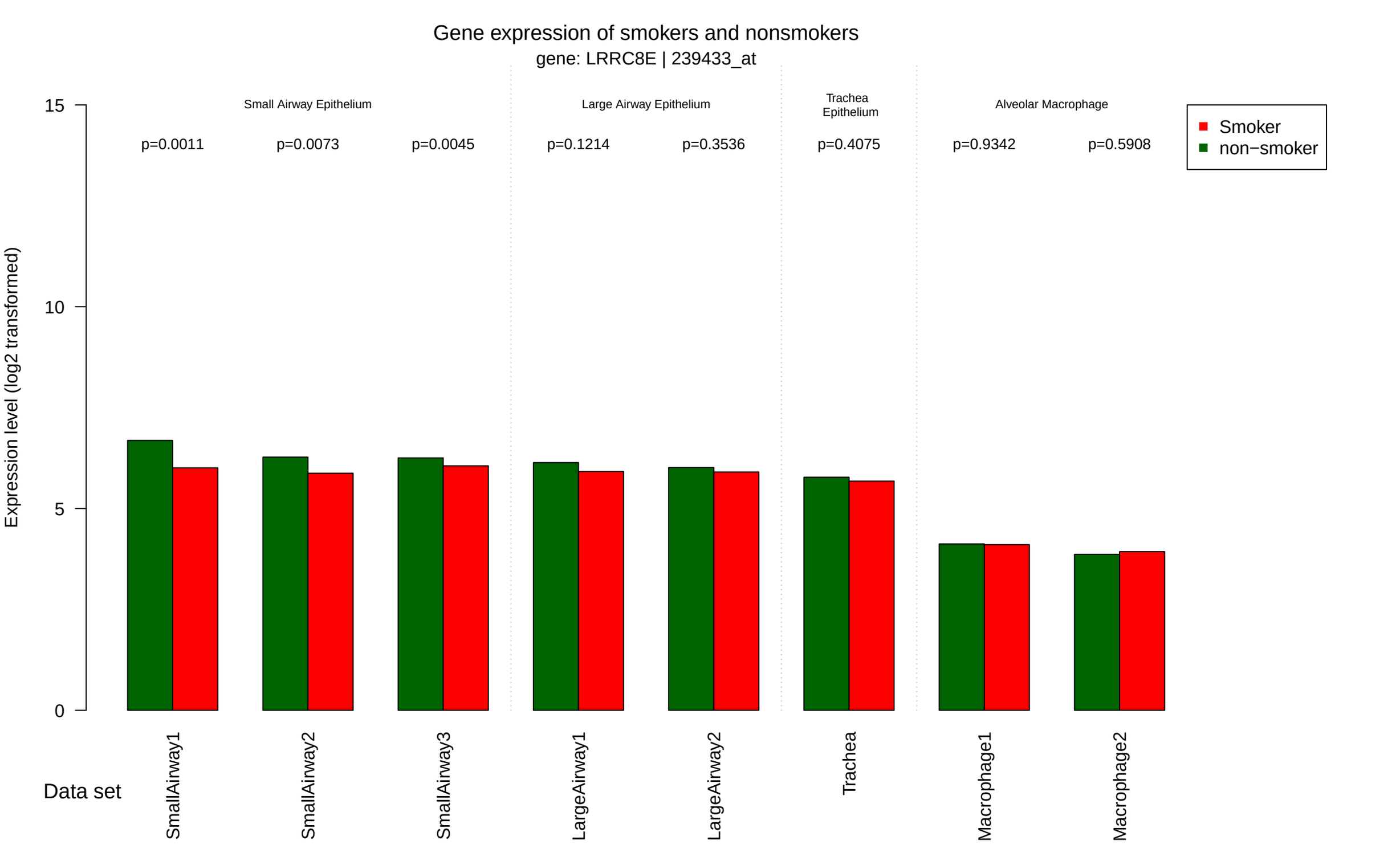
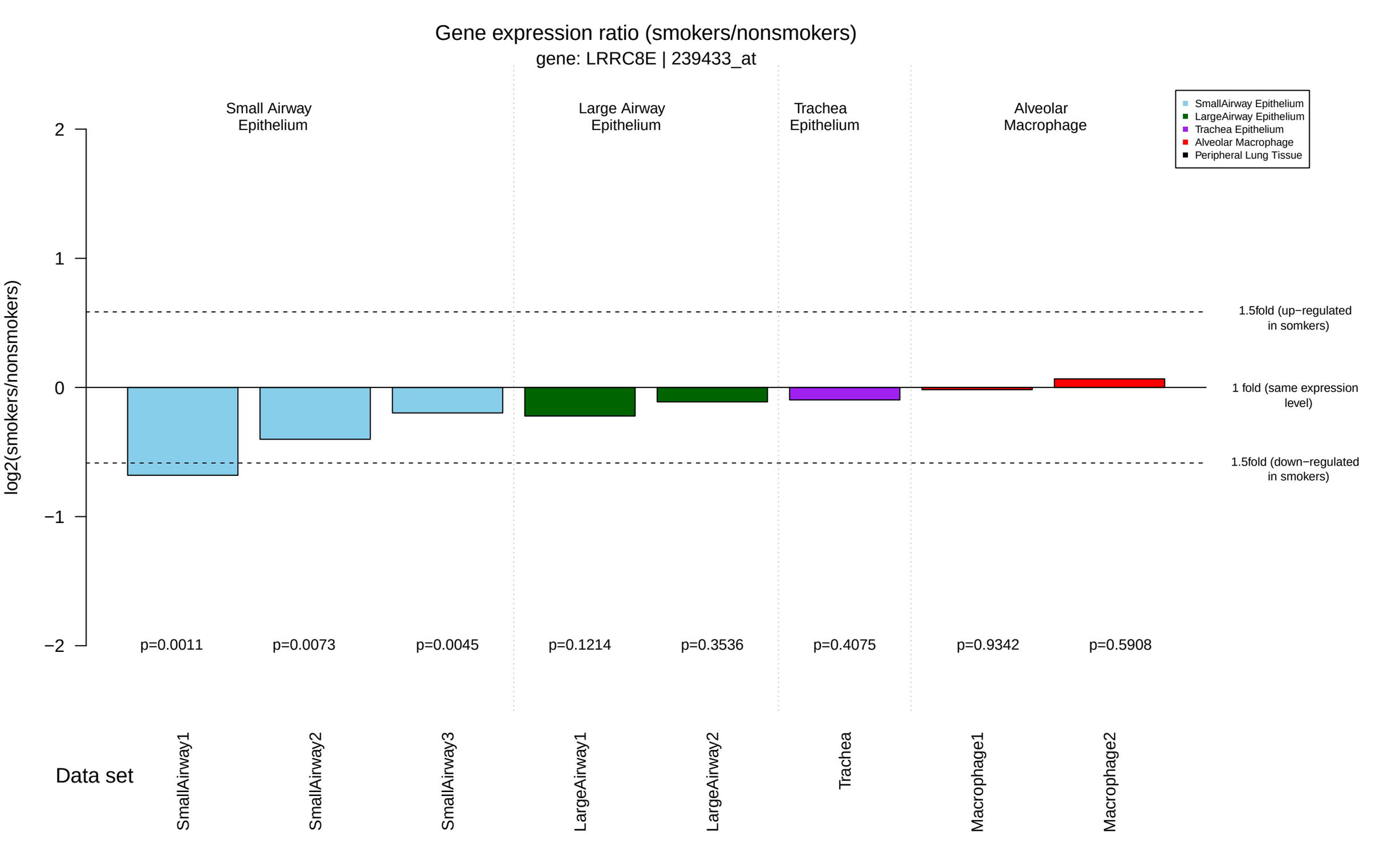
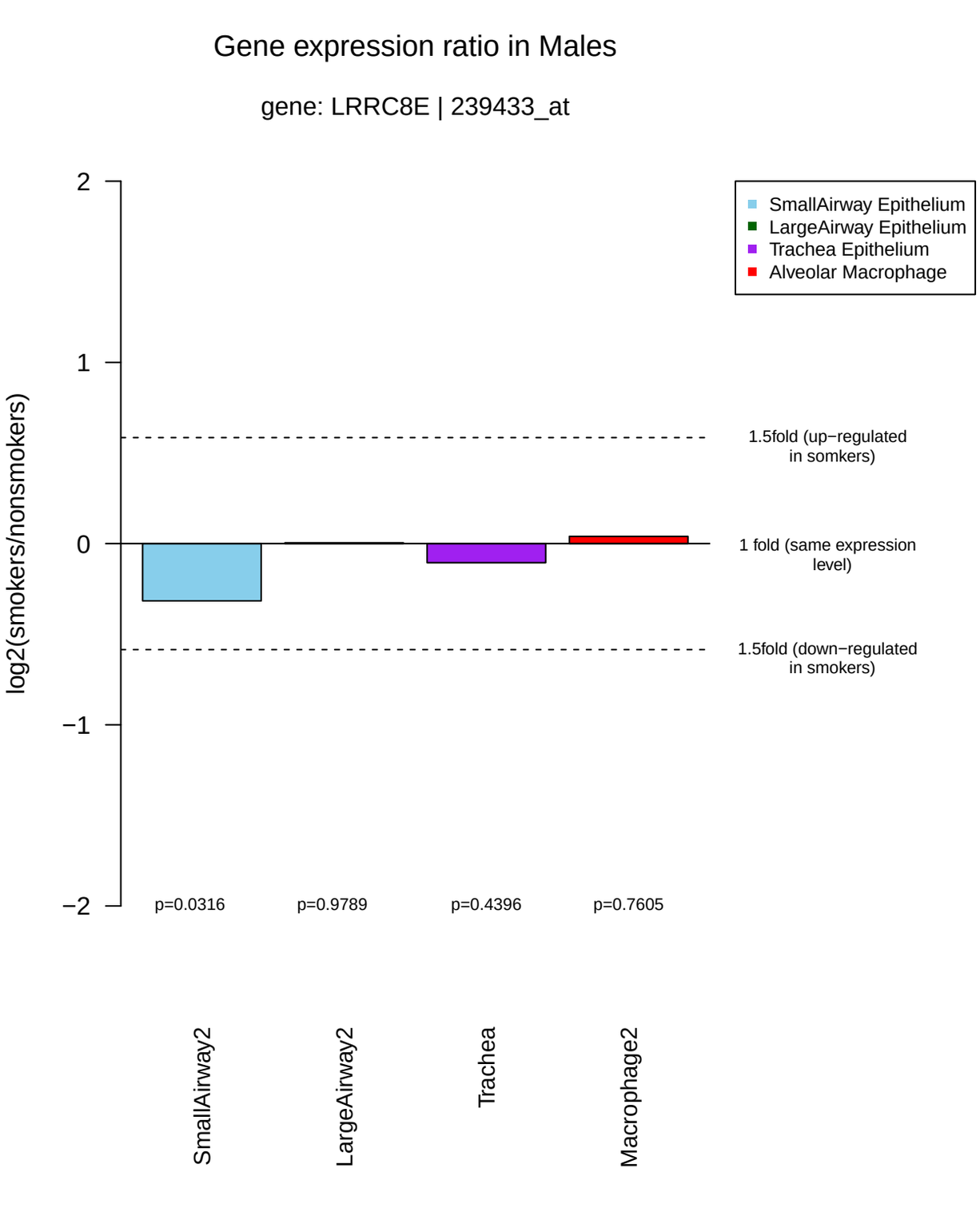
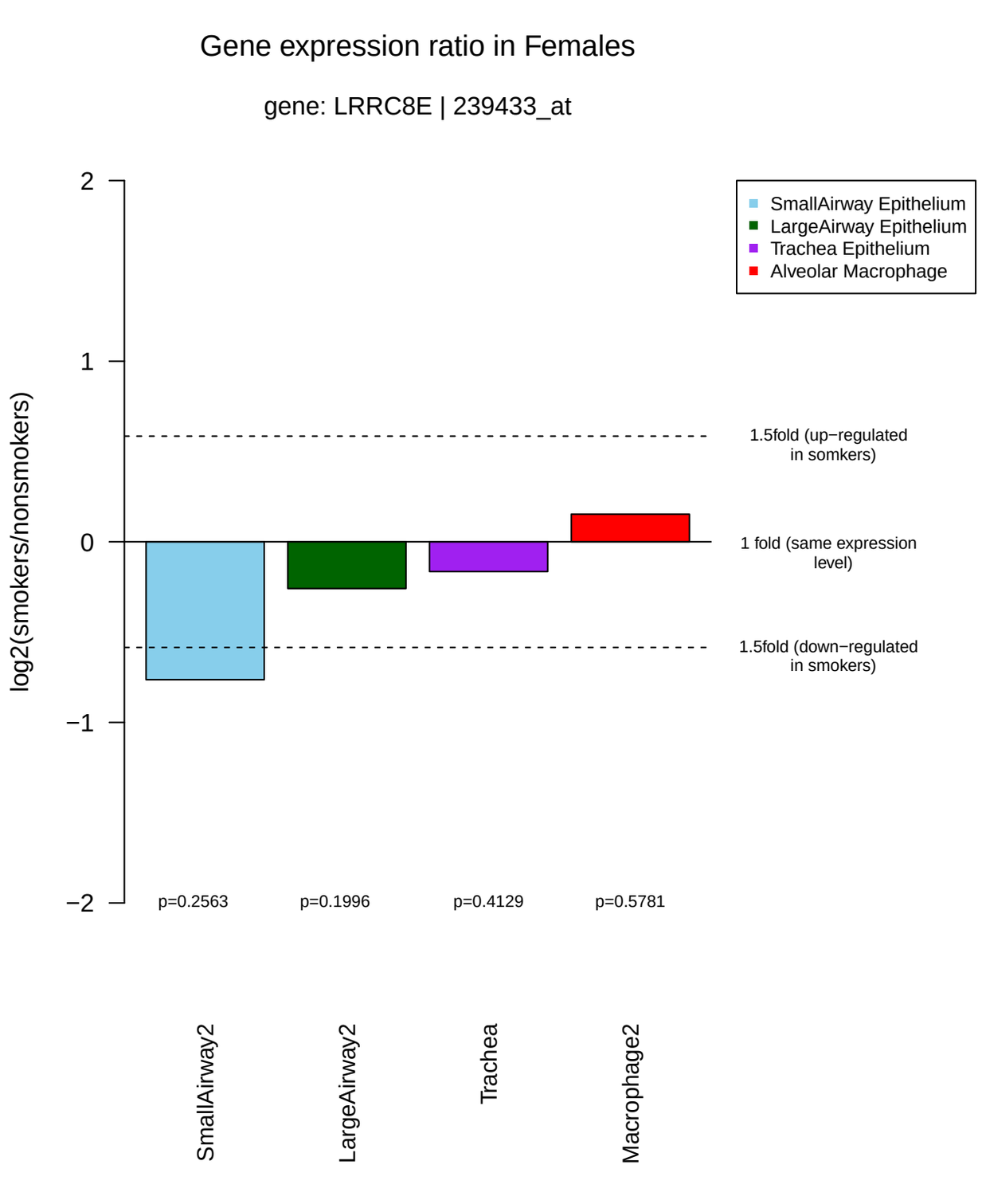
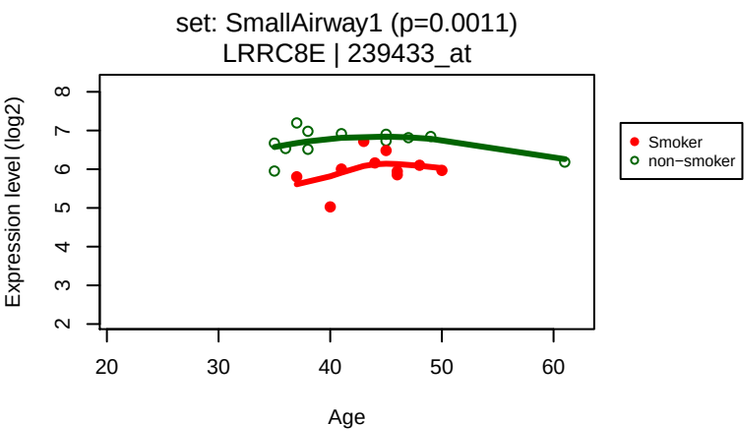
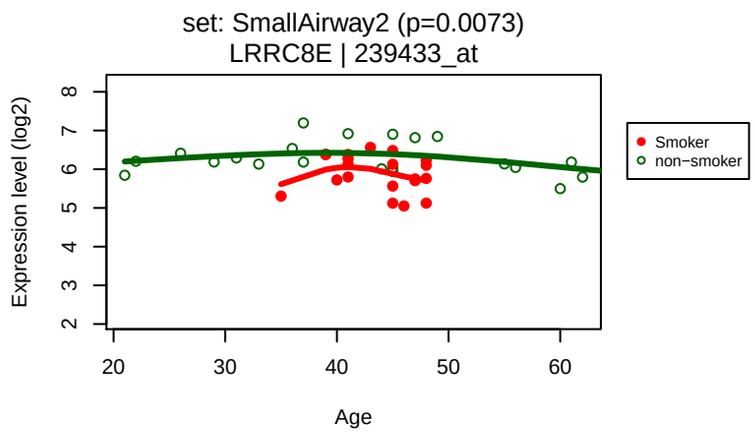

Gene expression of smokers and nonsmokers
gene: LRRC8E | 239433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0011
p=0.0073
p=0.0045
p=0.1214
p=0.3536
p=0.4075
p=0.9342
p=0.5908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

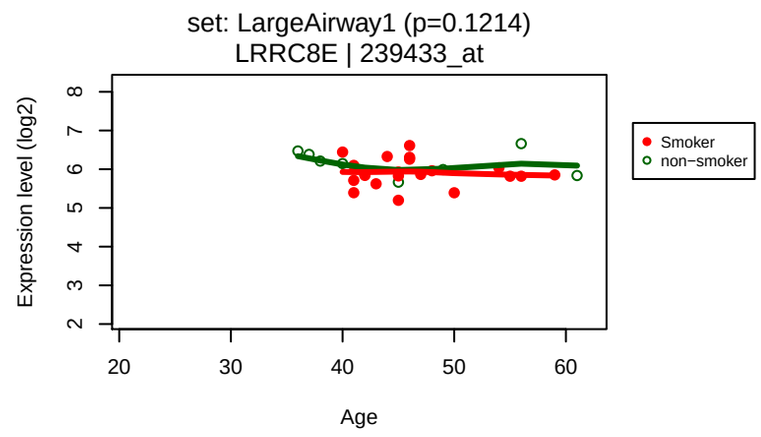

Gene expression ratio (smokers/nonsmokers)
gene: LRRC8E | 239433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0011
p=0.0073
p=0.0045
p=0.1214
p=0.3536
p=0.4075
p=0.9342
p=0.5908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

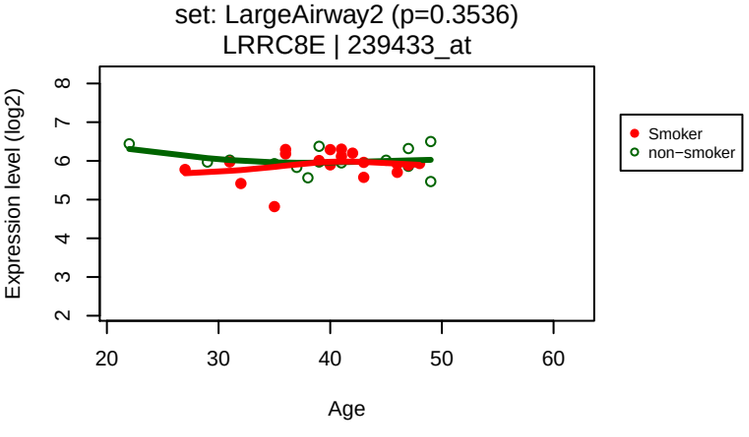

Gene expression ratio in Males
gene: LRRC8E | 239433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0316
p=0.9789
p=0.4396
p=0.7605
SmallAirway2
LargeAirway2
Trachea
Macrophage2

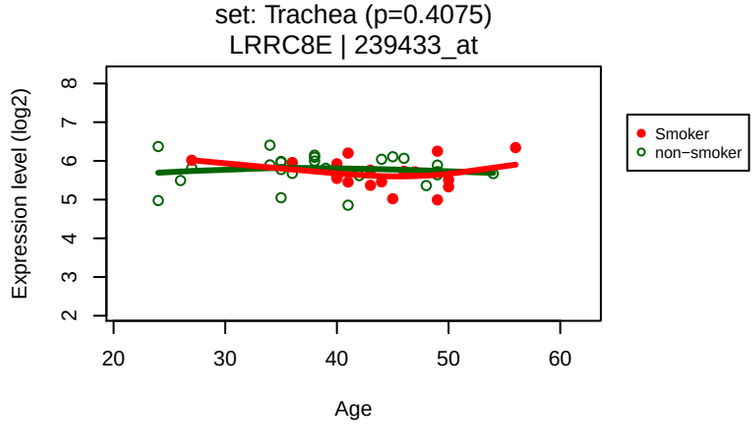

Gene expression ratio in Females
gene: LRRC8E | 239433_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2563
p=0.1996
p=0.4129
p=0.5781
SmallAirway2
LargeAirway2
Trachea
Macrophage2

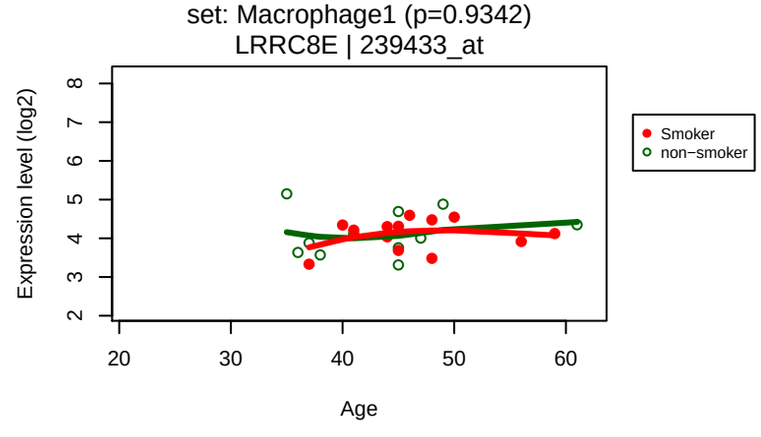

set: SmallAirway1 (p=0.0011)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker

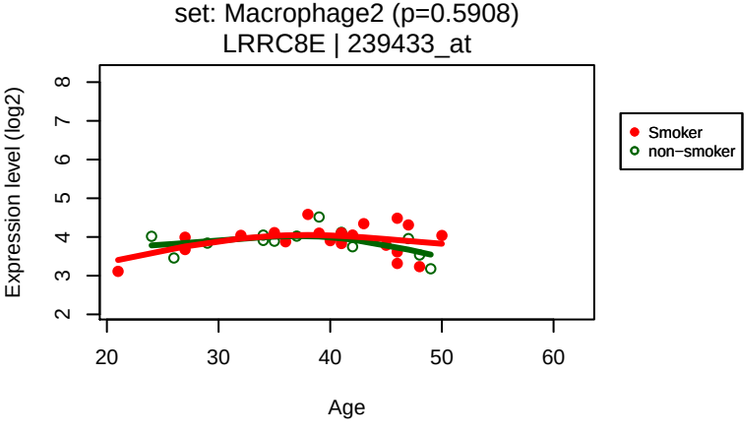

set: SmallAirway2 (p=0.0073)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker

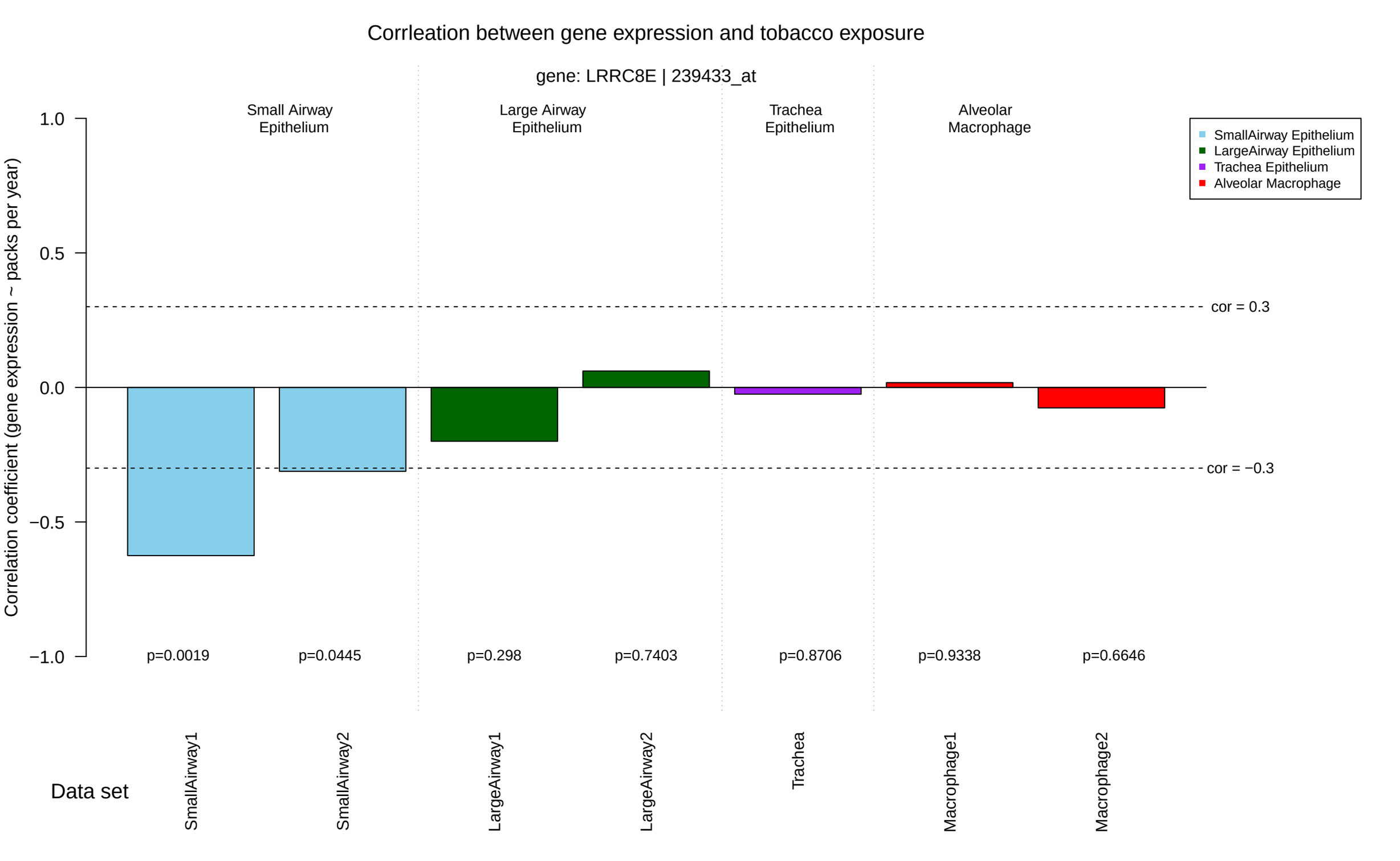

set: LargeAirway1 (p=0.1214)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3536)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4075)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9342)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5908)
LRRC8E | 239433_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LRRC8E | 239433_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0019
p=0.0445
p=0.298
p=0.7403
p=0.8706
p=0.9338
p=0.6646
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2